# MCQ A · integral, area and mean value

After S63; we always assume $a < b$.

For each question, only one option is correct. Just consult the solutions.

**1.** If $F' = f$ on $[a;b]$, then $\int_a^b f(x)\,\mathrm{d}x$ is:

A. $F(a) + F(b)$ B. $F(b) - F(a)$

C. $f(b) - f(a)$ D. $0$

**2.** The integral of a negative function on $[a;b]$ is:

A. positive B. zero

C. negative or zero D. always a positive area

**3.** The average value of $f$ on $[a;b]$ is:

A. $\frac{1}{b-a}\int_a^b f$ B. $\int_a^b f$

C. $f(a) + f(b)$ D. $b - a$

**4.** The geometric area between the curve and the x-axis requires:

A. preserving the signs B. split where the sign changes

C. a second derivative D. a random draw

**5.** An antiderivative of $e^x$ is:

A. $xe^x$ B. $e^x$

C. $\ln x$ D. $1/e^x$

## MCQ B · integration by parts and bounds

After S66.

For each question, only one option is correct. Justify your choice orally before consult the solutions.

**6.** For integration by parts, you must know:

A. two functions of class $\mathcal{C}^1$ on $[a;b]$ B. a geometric sequence

C. a fixed point D. a binomial distribution

**7.** $\int_a^b u'v\,\mathrm{d}x$ is equal to:

A. $[uv]_a^b - \int_a^b uv'\,\mathrm{d}x$ B. $\int_a^b uv\,\mathrm{d}x$

C. $[u+v]_a^b$ D. $0$

**8.** If $f$ is increasing and positive on $[0;1]$, left-endpoint rectangles give:

A. an upper bound B. a lower bound

C. always the exact integral D. a negative value

**9.** A sequence of integrals $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, with $f$ continuous and positive on $[0;1]$, may often be bounded using:

A. $0 \le x^n \le 1$ B. $x^n = n$

C. $\ln x \ge 0$ D. the derivative of $n$

**10.** A unit of the integral of a flow rate (L/min) over a time in min is:

A. $\mathrm{L/min^2}$ B. L

C. $\mathrm{min/L}$ D. without a unit

**True or false · justification expected.** For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. An integral is always positive.

2. Two antiderivatives of the same function on an interval differ by a constant.

3. A rectangle approximation removes the need to check the direction of the bounds.

**Review of prior work**

I can interpret sign, area and mean value; choose an antiderivative or integration by parts, and bound.

**To watch out for:** confusing geometric area with algebraic integral; forgetting the bounds in $[uv]_a^b$.